

Q1.

(a) Since $X_i \sim \text{Bernoulli}(0.5)$, $E(X_i) = 0.5$ & $\text{Var}(X_i) = 0.25$.
By linearity of expectation and independence;

$$E(S_n) = \sum_{i=1}^n E(X_i) = 0.5n = n/2$$

$$\text{Var}(S_n) = \sum_{i=1}^n \text{Var}(X_i) = 0.25n = n/4$$

(b) By defⁿ, standardizing a RV Y via $(Y - E(Y)) / \sqrt{\text{Var}(Y)}$ shifts its mean to 0 and scales its variance to 1. Substituting $E(S_n) = n/2$ and $\sqrt{\text{Var}(S_n)} = \sqrt{n/4}$ directly yields Z_n .

(c) For small n (like $n=10$), the distribution is visibly discrete and jagged. As n increases to 50 and 200, the gaps close up, asymmetry vanishes, and the histogram becomes a smooth, symmetric bell shaped curve matching the standard Gaussian density.

(d) CLT states that the normalized sum of a large number of independent, identically distributed (iid) RVs converges in distribution to a standard normal distribution $N(0,1)$, regardless of the underlying distribution. The convergence of the histograms and empirical parameters towards $N(0,1)$ shows this.

Monday	-	6	13	20	27
Tuesday	-	7	14	21	28
Wednesday	1	8	15	22	29
Thursday	2	9	16	23	30
Friday	3	10	17	24	31
Saturday	4	11	18	25	-
Sunday	5	12	19	26	-

08. week

Q2.

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• (a) By defⁿ, expected value of a fⁿ of a cont. RV $g(u)$ where $U \sim \text{Uniform}(0,1)$ is given by $\int_0^1 g(u) f_u(u) du$. Since the prob. density fⁿ $f_u(u) = 1$ on $[0,1]$, it reduces to,

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$$E[e^{-u^2}] = \int_0^1 e^{-u^2} (1) du \approx 1$$

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• (b) By the strong law of large numbers (SLLN), the sample average of iid RVs converges almost surely to the expected value (E) as the sample size $N \rightarrow \infty$. The statistical error scales at a rate of $O(1/\sqrt{N})$.

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• (c) Traditional grid based numerical integration methods (like the Trapezoidal rule) suffer from problems with dimensionality, where the computational complexity scales exponentially [$O(m^d)$ for d dimensions]. Monte Carlo integration's rate of convergence remains $O(1/\sqrt{N})$ independent of no. of dimensions ' d ', making it vastly superior for high-dimensional financial problems. (like multi-asset basket options).

2/2023				
Monday	-	6	13	20
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Q3.

(a) The probability of a uniformly distributed point falling inside a region is the ratio of the area of that region to the total area of sample space.

⇒ Area of Unit circle = π

⇒ Area of bounding square = $2 \times 2 = 4$

$$\text{Prob.} = \frac{\text{Area (circle)}}{\text{Area (square)}} = \frac{\pi}{4}$$

(b) This is also governed by law of large Numbers. The empirical fraction of points falling inside the circle serves as an unbiased estimator for the true probability ($\pi/4$). As $N \rightarrow \infty$, variance shrinks down, forcing the estimate closer to the actual value of π .

Q4.

(a) The terminal stock price S_T is a deterministic function of a standard normal RV Z ($S_T = S_0 e^{\sigma Z}$), meaning S_T is itself a RV (specifically following a log-normal distribution). Since the option pay off $(S_T - K)^+$ depends directly on S_T , it inherits this stochastic nature, making it a RV.

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⑧ Volatility affects outcomes as;

→ spread of terminal stock prices :- Higher volatility (σ) expands the width of the stock price distribution, resulting in a broader spread with fatter right-side tails.

→ Expected option payoff :- Higher volatility increases the expected payoff, because a call option's downside is truncated at 0 (we don't lose more if it finishes out of the money), it benefits entirely from the increased probability of large upwards movements on the right tail.

⑨ Many real-world derivatives possess paths or payoffs that don't have clean, closed-form analytical eq's (e.g.: Asian options tracking path averages, American choices, complex multi-asset logic). Monte Carlo lets us simulate thousands of realistic asset trajectories directly under the risk-neutral measure to value any arbitrary payoff structure easily.